

The Comprehensive Checklist for Building Robust, Explainable Customer Churn Prediction Systems

1. Data Acquisition and Quality Assessment

- **Identify Multi-Channel Sources:** Consolidate unstructured and structured data across all touchpoints, including CRM records, social media interactions, call center logs, service requests, chat messages, and mobile money transaction APIs.
- **Assess Churn Types:** Define specific target variables for the model based on business logic:
- **Active vs. Passive Churn:** Distinguish between explicit cancellations and non-renewal of subscriptions without communication.
- **Voluntary vs. Involuntary Churn:** Identify customers leaving due to dissatisfaction vs. payment failures (e.g., insufficient funds/expired cards).
- **Revenue vs. Customer Churn:** Separate the loss of total accounts from the loss of specific revenue streams (e.g., plan downgrades).
- **Execute Data Cleaning Protocol:**
- Remove constant features (e.g., Count, Country) that exhibit zero variance.
- Drop redundant identifiers and non-predictive labels like CustomerID and Phone Number.
- **RETAIN** geospatial features (Latitude, Longitude) and Churn Reason. *Note: Retain Churn Reason as a categorical feature despite high missingness (73%), as it is a primary driver in SHAP analysis.*
- Convert Total Charges and similar variables from object/string types to numeric formats, coercing errors to handle blank entries and feature sparsity.
- **Implement Imputation:**
- Apply **median imputation** for numerical features to mitigate the impact of right-skewed spending distributions and billing anomalies.
- Use **"most frequent" (mode) imputation** for categorical data to maintain the distribution of service usage and demographics.

2. Feature Engineering and Domain Rationale

- **Construct Advanced Metrics:**
- **AvgMonthlyCharge:** Generate indicator by dividing Total Charges by (Tenure + 1) to capture spending intensity relative to the customer lifecycle and identify "bill shock" risks.
- **HasMultipleServices:** Sum active service categories (Phone, Internet, Online Security, etc.) to quantify bundle complexity and switching costs.
- **Encoding Strategy:**
- Apply **Ordinal Encoding** to contract types (Month-to-month, 1-year, 2-year) to preserve the inherent hierarchy of customer commitment.
- Use **One-Hot Encoding** for non-ordinal categorical variables like payment methods or internet service types.

- **Geospatial Integration:** Treat **Latitude** and **Longitude** as scaled numeric inputs to allow the model to learn non-linear regional churn clusters and geographic service effects.

3. Class Imbalance and Leakage Prevention

- **Strict Leakage Control:**
- Split data into stratified training/validation/test sets (e.g., 80/20 or 64/16/20) **before** any oversampling is performed.
- Ensure the test set reflects real-world imbalance (approx. 26.5% churn) to provide an unbiased Brier Score and ROC-AUC.
- **SMOTE Implementation:** Address class imbalances in the training set using Synthetic Minority Over-sampling.
- Set the neighborhood size to **$k=5$** to balance local manifold structure with noise reduction.
- Apply SMOTE **exclusively to the training set** after the stratified split.

4. Iterative Model Development and Hyperparameter Tuning

- **Iteration 1 (Baseline):** Implement and benchmark a "Model Zoo" including **XGBoost, Random Forest, LightGBM, Gradient Boosting, AdaBoost, Logistic Regression, and Multi-layer Perceptron (MLP)**.
- **MLP Architecture:** Implement a 64-32-1 hidden layer structure using ReLU activation and L2 regularization (Alpha tuning).
- **Optuna-based Optimization:** Execute Bayesian optimization (50+ trials) to maximize **ROC-AUC**.
- Utilize a **3-fold stratified cross-validation** on the training set.
- Deploy a **Median Pruner** to handle trial efficiency and early stopping for poor-performing configurations.
- **Iteration 2 (SHAP-Driven Pruning):**
- Generate SHAP values for the top-performing baseline to identify global feature importance.
- Retain only the **top 20 features** to prune the feature space.
- **Target Metric:** Achieve a **~28% reduction in training time** while maintaining or improving validation AUC.

5. Advanced Ensemble and Stacking Architectures

- **Soft-Voting Ensemble:** Average predicted probabilities from the top-tier boosting models (**XGBoost, LightGBM, and Gradient Boosting**) to reduce prediction variance.
- **Stacking Framework:** Integrate five base classifiers (**Decision Tree, Random Forest, Extra Trees, Gradient Boosting, and XGBoost**) using a meta-classifier for final risk prediction.

6. Probability Calibration and Business Metrics

- **Calibration Protocol:**

- Apply **Isotonic Regression** for tree-based ensembles and the MLP to correct non-linear probability distortions.
- Use **Platt Scaling** specifically for Logistic Regression to ensure probabilities align with actual churn frequencies.
- **Cost-Sensitive Evaluation:** Define the business cost matrix using a **5:1 cost ratio**.
- **Cost of False Negative ($\$C_{\{FN\}} \$$) = 5:** Represents the cost of losing the customer (Customer Lifetime Value).
- **Cost of False Positive ($\$C_{\{FP\}} \$$) = 1:** Represents the cost of retention intervention (e.g., a \$20 discount/incentive).
- **Threshold Optimization:** Identify the specific optimal decision threshold of **0.528** to maximize the balance between precision and recall while minimizing total business costs.

7. Explainable AI (XAI) and SHAP Analysis

- **SHAP Explainer Selection:**
- Utilize TreeExplainer for all boosting and forest models for exact, efficient feature attribution.
- Deploy KernelExplainer for the MLP and Logistic Regression models, utilizing a **100-sample background set** for computational tractability.
- **Global Interpretability:** Rank features by mean absolute SHAP values (e.g., Contract type, Tenure, Churn Reason, Total day minutes).
- **Local Interpretability:** Use SHAP Force Plots and Waterfall Plots to decompose individual customer risk (e.g., explaining why a specific customer has an 87% churn probability).
- **Counterfactual Analysis:** Perform "What-If" simulations for retention.
- **Requirement:** Verify if converting a customer to a **1-year contract** successfully reduces churn probability by the **0.35 margin** identified in research.

8. Customer Segmentation and Targeted Retention

- **Latent Representation:** Use an **Autoencoder** with a symmetric encoder-decoder structure to learn **16-dimensional latent representations** of customer behavior.
- **Clustering Architecture:** Apply **K-Means (K=3)** to the latent space to discover high-utility segments.
- **Strategic Intervention Mapping:** Match segment profiles to prescriptive actions:
- **Electronic Check users:** Launch "Payment Method Migration" campaigns to move users to automated billing.
- **Fiber Optic users:** Initiate "Service Quality Reviews" to address technical churn drivers.
- **High-tenure/Price-sensitive users:** Deploy loyalty rewards and personalized bundle incentives.

9. Deployment and Market Interoperability

- **API and Open Platforms:** Ensure model integration with mobile money interoperability standards and wider financial ecosystems via open APIs.

- **Dashboarding:** Deploy multi-level visualizations for stakeholder tiers:
- **Executive Dashboards:** Aggregate SHAP summaries for strategic resource allocation.
- **Analyst Interaction Plots:** Detailed feature interaction and dependence plots for root-cause analysis.
- **ROI Monitoring:** Calculate financial impact by comparing the cost of interventions against preserved **Customer Lifetime Value (CLV)** and reduced customer acquisition costs.